

Teaching Statement

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Before my Ph.D., I spent two years as the instructor of record for seven undergraduate computer science courses at The Millennium University in Dhaka, where I was responsible for the syllabus, lectures, assessments, and final grades. At the University of Nevada, Reno, I then worked as a teaching assistant in four courses covering algorithms, machine learning, and large language models. I now supervise two thesis projects that I initiated at Khulna University while working as a postdoctoral researcher in cardiovascular AI. These experiences shaped the main principle of my teaching: students learn a concept best when they have to build something that depends on it.

Table 1: Teaching and mentoring record.

Role	Courses / students	Period
Lecturer (instructor of record)	Structured Programming; Data Structures; Database Systems; Digital Logic Design; Object-Oriented Analysis and Design; Software Engineering; Artificial Intelligence & Neural Networks; course coordinator	2019–2021
Graduate Teaching Assistant, UNR	CS 477/677 Analysis of Algorithms (4 semesters); CS 422/622 Introduction to Machine Learning; CS 491/691 Large Language Models and Multimodal AI; lab instructor, Programming for Data Science	2021–2026
Thesis supervisor, Khulna Univ.	Projects I initiated, co-supervised with Prof. G. M. Atiqur Rahman. M.S. thesis: pedestrian intent recognition for autonomous vehicles; B.S. thesis: coronary OCT segmentation	2026–present
Technical lead, industry	Four students (2 M.S., 2 B.S.) on ML system architecture and experiment design	2025–2026

Teaching approach

I organize courses around projects that students build and then test. In Object-Oriented Analysis and Design, students carried one project through the whole term, from requirements and UML models to design patterns and implementation. Design decisions made early in the term affected their work at the end, which made principles such as coupling and cohesion concrete. In algorithms and machine learning, I pair each concept with a short implementation exercise and a visual explanation of how the component fits into a larger system, and I ask students to explain their reasoning and compare alternative solutions. These frequent, low-stakes checks show me where the class is, and I adjust the pace and explanations accordingly.

Supporting students with different preparation

My students in Dhaka and Reno included first-generation, working, and non-traditional students as well as students with strong preparation. I use the same structure for both groups: clear expectations and rubrics, scaffolded assignments that build skills step by step, early low-stakes assessments that identify students who need help before the first exam, and optional extensions that give advanced students harder problems. In office hours, I focus on how a student is reasoning about a problem, so that they can solve the next one on their own.

Mentoring

My mentoring relies on early engagement, clear goals, and regular feedback. With my thesis students at Khulna University, who work in a different time zone, we agree on explicit milestones, and I ask for small early results, such as a reproduced baseline or an explained failure case, before we discuss new ideas. With the four students

I led in industry, I focused on engineering practice: clear architecture decisions, shared coding standards, and experiments planned before they are run. In my own lab, I will involve undergraduates early through well-defined subproblems, such as data curation, evaluation, or a single ablation study, that contribute to real projects and can lead to co-authorship.

Courses I can teach

Undergraduate: introductory programming (Python, C/C++), data structures, algorithms, database systems, artificial intelligence, machine learning, and introductory data science. **Graduate:** machine learning, deep learning, computer vision, medical image analysis, and large language models and multimodal AI. My teaching-assistant role in CS 491/691 prepares me directly for the last of these.

I would also develop two electives based on my research:

- **Plan, Activity, Intent, and Goal Recognition** (upper-division or graduate). Probabilistic, temporal, and deep learning models for inferring what agents are trying to do, with case studies in autonomous vehicles, maritime security, assistive technology, and human–robot collaboration. Prerequisite: introductory machine learning. Students complete a term project in which they build and evaluate a recognition module, including how early and how reliably it can commit to a prediction.
- **Deep Learning for Medical Image Analysis** (graduate or advanced undergraduate). Segmentation, detection, and multimodal fusion on clinical imaging data, with attention to label noise, inter-observer variability, calibration, and the evidence needed before a model should influence patient care. Students reproduce a published result and then test its limits.

Connecting research and teaching

I bring tools from my research into the classroom. Simulation environments such as NavySim let students test AI agents in controlled, repeatable scenarios and see how model errors affect decisions. Clinical imaging case studies show why accuracy alone is not enough when expert labels disagree and the stakes are high. In both settings, I ask students to report uncertainty and failure cases and to consider who is affected by a wrong prediction.

My goal is for students to leave my courses with solid foundations and the habit of checking their understanding by building and testing. Research and teaching support each other in my work, and I will keep them connected in my courses and my lab.